MANIFESTO FOR TECHNOLOGICAL SOVEREIGNTY

Independent strategic note & advocacy kit for energy, healthcare and agricultural autonomy.

THE CENTRAL IDEA

Produce energy and useful molecules locally from residual biomass, while creating durable carbon storage in solid form.

Designed for comfortable reading on both smartphones and computers.

01 — DIAGNOSIS

The Three Bottlenecks

01

Thermodynamic Risk

Heavy electrolysis competes directly with AI and global electrification. It dissipates anthropogenic heat without providing a carbon sink.

02

The Illusion of Solutions

Nuclear power and white hydrogen inject thermal energy into a CO_2-saturated system without providing carbon removal.

03

Financial Predation

Breakthrough SMEs can be weakened by short-selling and dilutive financing mechanisms (OCEANE/BSA) before industrial deployment.

02 — TECHNOLOGY

The Breakthrough: Thermolysis

A mature French technology requiring limited initial capital and ready for industrial deployment.

OBJECTIVE

Progressively replace oil with locally produced energy vectors, while valorising residual biomass and simultaneously creating a solid carbon sink.

- S-iC / C-iC cascade architecture
- Hydrogen, SAF and industrial syngas
- Solid biochar and CO_2 sequestration

The value comes from the cascade of products, not from a single energy vector.

03 — SYSTEM

The Cascade Architecture

BIOMASS

Organic waste
& agricultural residues

›

THERMOLYSIS

S-iC / C-iC modules
Cracking & valorisation

›

VECTORS

H_2 · SAF · Syngas
Industrial uses

›

CARBON

Solid biochar
CO_2 sequestration

A physical alternative to electrolysis: valorise local carbonaceous matter while storing part of the carbon in solid form.

The Figures Structuring the Model

50+

kg H_2/h — reference order of magnitude for the H6/C-iC configuration studied

500

kg H_2/h — performance communicated for the approximately 20 MW S-iC architecture

2.8

kWh of electricity per kg H_2 — figure retained for the S-iC

30

bar — delivery pressure used in the public LCOH reference

PUBLIC LCOH REFERENCE

Below €2.34/kg H_2 under the assumption of biomass at €90/t and electricity at €70/MWh. This value corresponds to hydrogen delivered at 30 bar.

The Biomass Lever

PUBLIC ASSUMPTION

669 kg/h of biomass at €90/t for a production of 50 kg H_2/h.

€60.21/h biomass

€1.204/kg H_2

MARGINAL SCENARIO — FREE BIOMASS

If the purchase cost of biomass disappeared while all other parameters remained unchanged: 2.34 − 1.204 = €1.136/kg H_2. Independent estimate, not a new manufacturer LCOH.

In a territorial model, the main costs then become collection, preparation/sorting, storage and transport.

Industrial Agility

Unlike centralised projects requiring complex building permits and heavy civil engineering, mobile modules redefine industrial agility.

15 days

deployment presented in the kit

0

foundation required

2–3 weeks

order of magnitude for commissioning

- Total mobility: modules can be moved according to available feedstock.
- Accelerated ROI: local operation and valorisation of several co-products.

Mobility: An Economic Example

BMW iX5 — COMPARISON PRESENTED IN THE KIT

Gasoline

≈ €10.00 / 100 km

6.5 L/100 km × €1.70/L

H_2

≈ €1.40 / 100 km

0.93 kg/100 km × €1.50/kg

≈ 7× difference in the example presented

Analysis based on 750 km of range for 7 kg of compressed H_2 distributed at €1.50/kg.

The Anti-Dilution Financial Shield

PEA-PME SAFE HAVEN

Protect French investors holding assets in PEA-PME accounts against organised predation.

BAN TOXIC OCEANE/BSA STRUCTURES

These mechanisms can destroy valuation and block bank credit.

SOVEREIGNTY ANTI-DILUTION FUND

Secure the capital of strategic energy champions before industrialisation.

STOP AGGRESSIVE SHORT-SELLING

Avoid financial pressure on strategic French technology companies.

Objective: Regulatory Fast-Track

LESS THAN 3 MONTHS

Reduce administrative review times for strategic mobile modules.

In the current kit, administrative inertia is presented as a barrier to innovation and deployment.

- Hospitals & healthcare centres
- Logistics hubs & heavy transport
- Agricultural cooperatives — biochar & SAF

URGENCY 2026

Enable local elected officials to quickly regain control over waste and energy policy.